



University of Piraeus
School of Information and Communication Technologies
Department of Digital Systems

Level: Master's Programme of Studies

Wireless and Mobile Network Security

Case 1

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Abstract

An understanding of the structure of the IEEE 802.11 protocol was achieved through the monitoring of wireless traffic, after first switching the TP-Link TL-WN722N adapter to Monitor Mode. Using Wireshark, frames were captured and classified into the three fundamental categories defined by the standard: Management, Control, and Data Frames.

Abstract

Understanding the structure of the IEEE 802.11 protocol was achieved by monitoring wireless traffic, after first successfully configuring the network interface card in Monitor Mode, enabling the capture of raw frames. Subsequently, by utilizing the Wireshark tool and applying targeted display filters, it was possible to identify and analyze the three main categories of the protocol. This process not only confirmed the theoretical architecture of wireless packets in practice, but also highlighted the absence of encryption and mandatory authentication in management messages, thus pinpointing the standard's vulnerability that served as the necessary foundation for the active attack in Use Case 2.

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Introduction

In this exercise, we will focus on capturing and analysing wireless network frames using the TP-Link TL-WN722N USB network adapter in Monitor Mode, within a Kali Linux environment.

The IEEE 802.11 standard defines the protocols for implementing Wireless Local Area Networks (WLAN) — what is commonly known as 'Wi-Fi'. Its architecture operates on two main layers:

- Physical Layer (PHY)
- Data Link Layer (DLL)

Monitoring at this low level is critical. If we examined higher-layer protocols such as HTTP (Layer 7), the wireless-specific mechanisms would be invisible. By monitoring at the 802.11 layer, the complete communication logic of the wireless environment becomes visible.

According to the theory, the 802.11 frames we will search for and analyse are divided into three main categories:

- **Control Messages:** They facilitate the exchange of data frames between stations. Characteristic examples are the mechanisms of frame acknowledgement (ACK) and the Request to Send / Clear to Send (RTS/CTS).
- **Management Messages:** They allow the maintenance or termination of communication (e.g. Beacon frames broadcast by routers to announce their presence, and Deauthentication frames used to terminate connections).
- **Data Messages:** These are the frames that carry the actual payload, such as packets from websites, files, etc.

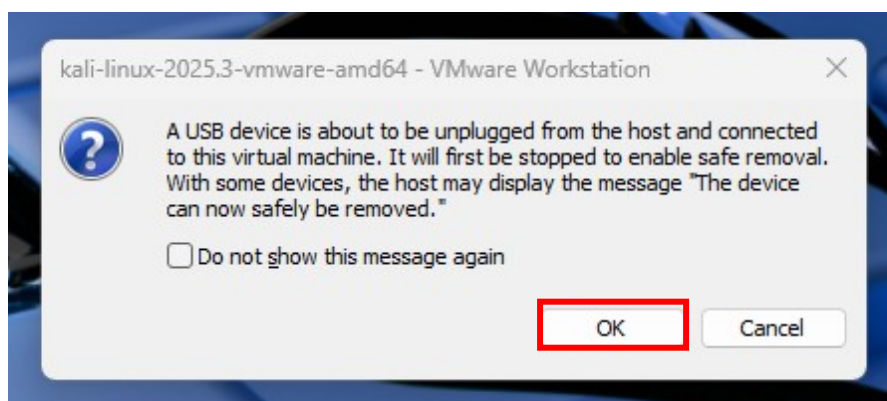
As a natural continuation of the above analysis, in Use Case 2 we will proceed to the practical exploitation of the protocol's weaknesses, specifically through a Deauthentication attack.

Practical part of the assignment

Regarding the first step of connecting the equipment to VMware, the process begins by inserting the TP-Link TL-WN722N wireless USB adapter into a free USB port of the laptop, then navigating to 'VM > Removable Devices' in VMware and connecting the card exclusively to Kali Linux.

When the device is connected, VMware displays a standard informational message confirming that the TP-Link card has been released from the host machine and attached exclusively to the Kali Linux virtual machine.

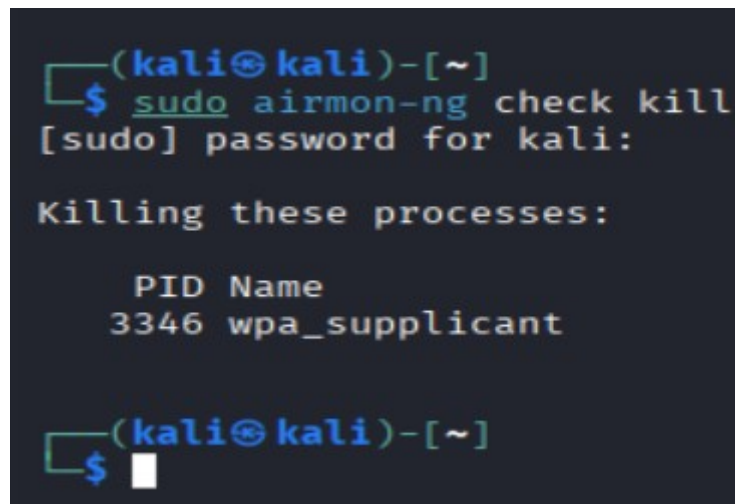
In
to
the



order
verify

successful recognition of the wireless interface by the operating system, a terminal is launched within Kali Linux and the `iwconfig` command is executed.

As shown by the command results, Kali Linux successfully recognised the external network adapter, assigning it the interface name wlan0. The card's default operating mode was 'Managed'. The command `sudo airmon-ng check kill` was then executed to terminate background services that could interfere with Monitor Mode.



```
(kali㉿kali)-[~]  
$ sudo airmon-ng check kill  
[sudo] password for kali:  
  
Killing these processes:  
  
PID Name  
3346 wpa_supplicant  
  
(kali㉿kali)-[~]  
$
```

During the execution of the previous command, the system successfully detected and terminated the `wpa_supplicant` process. The wireless card was then switched to Monitor Mode via `sudo airmon-ng start wlan0`.

As confirmed by the results, the interface's operating mode changed successfully to Monitor Mode.

In order to isolate the desired traffic from the overall network noise and classify the collected frames, Wireshark was used with the three fundamental type filters.

First, using the filter `wlan.fc.type == 0`, Management Frames were isolated — these govern the wireless network infrastructure (connection establishment, network announcements, disconnection).

Subsequently, the filter `wlan.fc.type == 1` allowed the exclusive display of Control Frames, in which the ACK (Acknowledgement) mechanism was confirmed.

Finally, via the filter `wlan.fc.type == 2`, Data Frames were revealed — these carry the actual packet payload.

Applying this analysis methodology, packet number 6 in the recording was examined in detail — a Management Frame.

The in-depth analysis of the 'Frame Control Field' fully verified the theoretical structure of the 802.11 protocol. Specifically:

- Protocol Version: Value 0 recorded, corresponding to the current protocol version.
- Frame Type: Value 0 (00 in binary) detected, certifying the packet's character as a Management message.
- Subtype: Value 8 recorded, identifying this specific frame as a Beacon message.

Beyond the frame categorisation, examining the routing parameters is of particular interest. The Source Address and BSSID fields contain the MAC address of the router, while the Destination Address carries the broadcast address (FF:FF:FF:FF:FF:FF). With the filter 'wlan.fc.type == 1', Wireshark successfully isolated the Control Frames. Analysis of the selected packet revealed:

- Frame Type: Value 1 (01 in binary) recorded, certifying that the packet belongs to the Control message category.
- Subtype: Value 13 recorded, which according to the 802.11 standard corresponds exclusively to ACK (Acknowledgement) messages.

These findings fully align with the theoretical framework of wireless network operation. According to the theory, every data frame must be explicitly acknowledged by the receiving station.

By applying the filter 'wlan.fc.type == 2', Data Frames were successfully isolated. Analysis revealed:

- Frame Type: Value 2 (10 in binary) recorded, confirming the packet's membership in the third main category.
- Subtype: Value 0 recorded, indicating a standard Data frame.

The above findings fully match the theoretical background of the 802.11 standard. As provided by the theory, data frames are the carriers of the actual information payload and are protected by WPA2 encryption.

Conclusion

Completing the first stage of the exercise, practical monitoring and analysis of wireless traffic at the MAC layer was achieved, with three key categories of IEEE 802.11 frames identified and verified.

- Management Frames (Type 0): The existence of Beacon frames (Subtype 8) was confirmed, broadcast freely to announce the network.
- Control Frames (Type 1): ACK (Subtype 13) messages were recognised, essential for reliable data transfer.
- Data Frames (Type 2): Data frames (Subtype 0) were identified, exclusively responsible for carrying the actual payload.

The in-depth understanding of the structure of these packets is not merely a theoretical exercise, but the foundation of the offensive and defensive methodologies examined in the subsequent use cases.

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